Decoupled Phase Space Flexing in Autonomous Hamiltonian Systems

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***Abstract*—Hamiltonian dynamical systems can be studied with the tools of symplectic geometry, which constrain how volumes of space deform. A better understanding of phase space flexing outlines how control strategies can leverage natural deformations to reduce control effort. This letter establishes analytical forecasts for the stretching of phase space in the vicinity of a reference trajectory. For autonomous Hamiltonian systems, we leverage symplectic invariants and exterior algebraic forms to predict the expansion and contraction of phase space along both the local flow direction and the local gradient direction. This isolates an area-preserving sub-pocket of phase space spanned by these two directions. The stretching results are demonstrated numerically and linked to control effort in the two-body problem.**

***Index Terms*—Aerospace, algebraic/geometric methods, autonomous systems.**

I. Introduction

WITHOUT a sufficient propellant budget or an appropriate control strategy, a spacecraft may deviate from its desired path, potentially leading to mission failure. To characterize control effort, fuel-minimizing control strategies have been developed to guide spacecraft into target orbits [1]. Moreover, Monte Carlo simulations are commonly used to provide robust estimates of state deviations and related maneuver magnitudes required for insertion into target orbits [2]. However, these descriptions of control effort involve numerically sensitive boundary value problems and computationally expensive simulations.

This letter gives analytical insight into phase space deviations in Hamiltonian dynamical systems, which have implications for control effort. We leverage the geometric structure of autonomous Hamiltonian systems to describe the local expansion and contraction of phase space [3]. The separation between neighbouring trajectories fluctuates according to the deformation of the local phase space. Control strategies that aim to drive these separations to zero require more effort when they oppose the natural dynamical flow. Extending this concept, control strategies can collapse volumes of phase space onto a reference trajectory, where these volumes characterize uncertainty in spacecraft position and velocity. Efficient control strategies exploit the natural deformation of these volumes to reduce control effort. Thus, the local flexing of phase space volumes informs how fuel-efficient control efforts respond to state uncertainty under the dynamics.

Optimal control problems are also Hamiltonian with an extended phase space, introducing canonical costate variables that characterize the control effort [4], [5]. Then, the flexing of the extended phase space directly informs the evolution of costates and, consequently, control effort. Additionally, generalized Hamiltonian systems, such as port-Hamiltonian systems, have been applied to the control of spacecraft formations [6]. Understanding phase space flexing in simpler, autonomous systems may serve as a foundation for extending these ideas to more complex systems [7].

Previous studies have examined the local flexing of phase space. Finite-time Lyapunov exponents characterize the local stability of regions of space and determine whether neighbouring trajectories diverge or converge over a finite time horizon [8]. The linearized dynamical flow has also been used to identify pinching and stretching directions around a reference state through an eigendecomposition [9], [10]. However, these studies require numerical integration of the local space surrounding a state in order to determine its stretching behaviour. Thus, our control problem seeks to uncover stretching information without numerical integration.

Our insight into phase space stretching and the corresponding response of efficient control strategies is built using natural dynamical invariants induced by the symplectic structure inherent in Hamiltonian systems [11]. These invariants preserve measures of space, specifically volumes and areas. We explicitly demonstrate the preservation of volume-like invariants in subspaces that compose a neighbourhood of space around a reference state. This provides predictable phase space stretching forecasts in these subspaces. In one subspace, the local flexing completely matches local changes in the Hamiltonian value. These subspaces are decoupled from one another, such that stretching or pinching of space along one basis vector in a subspace induces no corresponding stretching along basis vectors in other subspaces.

As shown in Theorem 3 (*Decoupled Phase Space Stretching*), the main contribution of this letter is the decomposition of phase space about a reference state into two

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decoupled subspaces. Phase space flexing and control effort in the lower-dimensional subspace is predictable without the need for the numerical integration. Section II outlines concepts relevant for a geometric description of phase space. Section III presents the control problem tied to the local flexing of phase space. Our analytical decomposition is presented in Section IV. Section V demonstrates a numerical example in the two-body problem.

II. Background

To understand phase space stretching, we first present preliminary information for Hamiltonian systems and exterior algebraic forms.

A. Hamiltonian Systems

The motion of a particle subjected to non-dissipative forces is Hamiltonian. Hamiltonian mechanics places the dynamical evolution of particles in terms of canonical positions (q_i) and canonical momenta (p_i).

Definition 1 (Hamilton's Equations): An n degree-of-freedom dynamical system is said to be Hamiltonian if the canonical coordinate equations of motion are generated by a scalar function $\mathcal{H}$. Defining the state $\mathbf{z} \in \mathbb{R}^{2n}$ such that $\mathbf{z} = (q_1, \dots, q_n, p_1, \dots, p_n)$, the equations of motion are provided by Hamilton's equations,

$$\frac{\mathrm{d}\mathbf{z}}{\mathrm{d}t} = -J\nabla_z \mathcal{H}(\mathbf{z}) \text{ where } J = \begin{bmatrix} \mathbb{O}_{n\times n} & -\mathbb{I}_{n\times n} \\ \mathbb{I}_{n\times n} & \mathbb{O}_{n\times n} \end{bmatrix} \tag{1}$$

Note that $\mathcal{H}$ denotes the Hamiltonian of the dynamical system, and that $J^T = -J$ and $JJ = -\mathbb{I}_{2n\times 2n}$.

The solution to (1), known as the solution flow (ϕ_t), induces symplectic geometry in phase space where measures of space, such as volume and area, are then geometrically constrained. For example, a $2n$-dimensional region of phase space must preserve its $2n$-dimensional volume, regardless of shape changes. We can then leverage these invariant quantities from symplectic geometry to constrain motion.

Definition 2 (Linearized Hamilton's Equations): The equations of motion in (1) can be linearized about a reference state $\mathbf{z}_0 \in \mathbb{R}^{2n}$ to describe neighbouring motion. We can define $\delta\mathbf{z}_0 = \mathbf{z}_0' - \mathbf{z}_0$ which can be identified with a vector in the tangent space $\mathcal{T}_{z_0}\mathbb{R}^{2n}$ about $\mathbf{z}_0$. The dynamics governing $\delta\mathbf{z}_0$ are given as,

$$\frac{\mathrm{d}}{\mathrm{d}t}(\delta\mathbf{z}_0) = -J\nabla_{zz}\mathcal{H}(\mathbf{z}_0)\delta\mathbf{z}_0, \tag{2}$$

where $\nabla_{zz}\mathcal{H}$ is the Hessian matrix of $\mathcal{H}$.

The state transition matrix (Φ) is the solution to (2) and is also a linearization of the solution flow (ϕ_t). The matrix Φ can be viewed as an operator mapping the tangent space about the point $\mathbf{z}_0$ to the tangent space about the point $\phi_t(\mathbf{z}_0)$. To this end, Ref [12] demonstrates that the vector, $\delta\mathbf{z}_0$, residing in the tangent space $\mathcal{T}_{z_0}\mathbb{R}^{2n}$ also evolves according to Hamilton's equations (1). In other words, $\delta\mathbf{z}_0$ is associated with a canonical coordinate set. Since the state transition matrix (Φ) is independent of $\delta\mathbf{z}_0$, Φ can be treated as a constant, symplectic matrix when mapping $\delta\mathbf{z}_0$. The resulting mapped coordinates $\delta\mathbf{z} = \Phi\delta\mathbf{z}_0$ are then also canonical [13].

B. Exterior Algebraic Forms

To continue on, we must introduce notation and concepts from exterior algebraic forms [11]. An exterior algebraic form is a function that maps a collection of vectors to a scalar quantity, often related to geometric information of the collection of vectors. The simplest example is a 1-form. We can define two functions $f_1, f_2 : \mathbb{R}^{2n} \to \mathbb{R}$ and refer to them as 1-forms because they take a single vector as input and return a scalar quantity. That is, $f_1(\xi) = \alpha$ and $f_2(\xi) = \beta$. From 1-forms, more complicated k-forms can be generated.

Definition 3 (Exterior Product): The exterior product is a 2-form, taking in two vectors $\xi_1, \xi_2 \in \mathbb{R}^{2n}$ as input and returning a scalar. It is defined using 1-forms,

$$f_1 \wedge f_2(\xi_1, \xi_2) = \det\left(\begin{bmatrix} f_1(\xi_1) & f_2(\xi_1) \\ f_1(\xi_2) & f_2(\xi_2) \end{bmatrix}\right). \tag{3}$$

The exterior product is bilinear and skew-symmetric.

In this letter, we are interested in differential, coordinate 1-forms $\mathrm{d}z_i : \mathcal{T}_z\mathbb{R}^{2n} \to \mathbb{R}$ which take a vector in the tangent space of our phase space as input and extract the variation of the z_i-coordinate along such a vector. Following (3), differential, coordinate 1-forms can be used to construct exterior products. For example, if $\xi_1, \xi_2 \in \mathcal{T}_z\mathbb{R}^2 \sim \mathbb{R}^2$, then $\mathrm{d}p \wedge \mathrm{d}q(\xi_1, \xi_2)$ returns the area of the parallelogram spanned by ξ_1, ξ_2 in a 2-dimensional phase space. This generalizes to higher dimensional phase spaces where exterior monomials, "chains" of exterior products, are constructed to compute higher dimensional geometric quantities. For instance, $\mathrm{d}p_1 \wedge \dots \wedge \mathrm{d}p_n \wedge \mathrm{d}q_1 \wedge \dots \wedge \mathrm{d}q_n(\xi_1, \dots, \xi_{2n})$ represents $2n$-dimensional volume spanned by the tangent space vectors $\xi_1, \dots, \xi_{2n}$. In the upcoming analysis, we rewrite $\mathrm{d}p_1 \wedge \dots \wedge \mathrm{d}p_n \wedge \mathrm{d}q_1 \wedge \dots \wedge \mathrm{d}q_n(\xi_1, \dots, \xi_{2n}) \to \mathrm{d}p_1 \wedge \dots \wedge \mathrm{d}p_n \wedge \mathrm{d}q_1 \wedge \dots \wedge \mathrm{d}q_n$, where it is implied the differential forms are acting on tangent vectors stemming from some point in phase space.

III. Problem Formulation

Consider an n degree-of-freedom, autonomous Hamiltonian system. The full particle state is given by $\mathbf{z} = (q_1, \dots, q_n, p_1, \dots, p_n)$ where the time rate is given in (1). The phase space then forms a symplectic manifold $(\mathbb{R}^{2n}, \omega)$ where ω is the standard symplectic 2-form.

We then initialize an infinitesimal, $2n$-dimensional volume element about a reference state $\mathbf{z}_0 \in \mathbb{R}^{2n}$ with sidelengths $\mathbf{u}_i, \mathbf{v}_i \in \mathcal{T}_{z_0}\mathbb{R}^{2n}$. Through the flexing of these sidelengths, we aim to analytically understand shape deformations of the volume element, which informs how spacecraft-guiding control strategies can be designed in a fuel-minimizing manner. First, we need to use the invariant structure captured by the standard symplectic 2-form.

Definition 4 (Standard Symplectic 2-Form): The standard symplectic 2-form ω is induced by a phase space $\mathcal{M} = \mathbb{R}^{2n}$. Together, the phase space and symplectic 2-form define our symplectic manifold of the form $(\mathbb{R}^{2n}, \omega)$. For this symplectic manifold and any $\xi_1, \xi_2 \in \mathcal{T}_z\mathbb{R}^{2n}$, a global, matrix representation for ω can be given as,

$$\omega(\xi_1, \xi_2) = \sum_{i=1}^{n} \mathrm{d}p_i \wedge \mathrm{d}q_i(\xi_1, \xi_2) = -\xi_1^T J\xi_2. \tag{4}$$

At any point on our symplectic manifold $(\mathbb{R}^{2n}, \omega)$, a local symplectic basis for the tangent space can be constructed. This basis spans the tangent space, with n basis vectors $\mathbf{u}_i$ associated with canonical positions q_i and n basis vectors $\mathbf{v}_i$ tied to canonical momenta p_i.

Definition 5 (Symplectic Basis): A symplectic basis B for the tangent space $\mathcal{T}_z\mathbb{R}^{2n}$ is a set of $2n$ linearly independent vectors where $B = \{\mathbf{u}_1, \ldots, \mathbf{u}_n;\ \mathbf{v}_1, \ldots, \mathbf{v}_n\}$. Additionally, these basis vectors must satisfy,

$$\omega\big(\mathbf{u}_i, \mathbf{u}_j\big) = \omega\big(\mathbf{v}_i, \mathbf{v}_j\big) = 0, \quad \omega\big(\mathbf{u}_i, \mathbf{v}_j\big) = \delta_{ij} \quad \forall\, i, j = 1, 2, \ldots, n. \tag{5}$$

If the basis vectors are also orthogonal, then the symplectic basis B is referred to as an orthosymplectic basis.

An orthosymplectic basis can be used to construct a change of basis matrix, $R(\mathbf{z_0})$, which is grounded to a point $\mathbf{z}_0$ on the symplectic manifold $(\mathbb{R}^{2n}, \omega)$ and independent of $\delta\mathbf{z}_0$. The matrix $R(\mathbf{z_0})$ can also map the tangent space coordinates δz_0. Stemming from an orthosymplectic basis, $R(\mathbf{z_0})$ is then also a constant, symplectic matrix. As with the state transition matrix (Φ), the resulting coordinates $R(\mathbf{z_0})\delta z_0$ are canonical and obey Hamilton's equations [13].

IV. Theory

We first need to identify an appropriate orthosymplectic basis to decompose the tangent space about a reference state.

Theorem 1 (Flow-Gradient Orthosymplectic Basis) : The flow direction $-J\widehat{\nabla_z\mathcal{H}}(\mathbf{z_0})$ and the gradient direction $\widehat{\nabla_z\mathcal{H}}(\mathbf{z_0})$ are members of an orthosymplectic basis spanning the tangent space $\mathcal{T}_{z_0}\mathbb{R}^{2n}$ of a symplectic manifold $(\mathbb{R}^{2n}, \omega)$ for an autonomous Hamiltonian system.

Proof: For this proof, we develop an orthosymplectic generalization of the Gram-Schmidt process. Rather than to generate a series of $2n$ orthogonal vectors as with the traditional Gram-Schmidt process, we generate bases for a Lagrangian subspace of $\mathcal{T}_{z_0}\mathbb{R}^{2n}$ and its complement. Together, the bases form an orthosymplectic basis for $\mathcal{T}_{z_0}\mathbb{R}^{2n}$. An n-dimensional Lagrangian subspace U is a subspace whose vectors $\mathbf{u}_i \in U$ produce a zero symplectic 2-form, $\omega(\mathbf{u}_i, \mathbf{u}_j) = 0,\ \forall \mathbf{u}_i, \mathbf{u}_j \in U$. Its complement is then another n-dimensional subspace V satisfying $\omega(\mathbf{v}_i, \mathbf{v}_j) = 0,\ \forall \mathbf{v}_i, \mathbf{v}_j \in V$ and $\omega(\mathbf{u}_i, \mathbf{v}_j) = \delta_{ij},\ \forall \mathbf{u}_i \in U, \mathbf{v}_j \in V$.

Consider n linearly independent vectors $\{\tilde{\mathbf{u}}_i\}_{i=1}^n$. These vectors are members of the tangent space $\mathcal{T}_{z_0}\mathbb{R}^{2n}$ with coordinate representations belonging to $\mathbb{R}^{2n}$. We first show the vectors $\{\tilde{\mathbf{u}}_i\}_{i=1}^n$ can be transformed into an orthogonal basis for an n-dimensional Lagrangian subspace of $\mathcal{T}_{z_0}\mathbb{R}^{2n}$. Subsequent basis vectors are constructed such that they do not lie along previous basis vectors nor in the complement of the Lagrangian subspace. The recursive algorithm,

$$\begin{aligned}
\mathbf{u}_1 &= \tilde{\mathbf{u}}_1\\
\mathbf{u}_2 &= \tilde{\mathbf{u}}_2 - \left(\frac{\tilde{\mathbf{u}}_2^T\mathbf{u}_1}{||\mathbf{u}_1||}\right)\mathbf{u}_1 - \left(\frac{\tilde{\mathbf{u}}_2^T J\mathbf{u}_1}{||J\mathbf{u}_1||}\right)J\mathbf{u}_1\\
&\cdots\\
\mathbf{u}_j &= \tilde{\mathbf{u}}_j - \sum_{k=1}^{j-1}\left[\left(\frac{\tilde{\mathbf{u}}_j^T\mathbf{u}_k}{||\mathbf{u}_k||}\right)\mathbf{u}_k - \left(\frac{\tilde{\mathbf{u}}_j^T J\mathbf{u}_k}{||J\mathbf{u}_k||}\right)J\mathbf{u}_k\right]
\end{aligned} \tag{6}$$

with $\mathbf{u}_j \to \mathbf{u}_j/||\mathbf{u}_j||$, generates a series of unit length vectors $\{\mathbf{u}_i\}_{i=1}^n$ that form an orthogonal basis for an n-dimensional Lagrangian subspace. To show this, we proceed with an induction based proof. Starting with the base case, we consider $\mathbf{u}_1$ and $\mathbf{u}_2$ in (6). Note that,

$$\begin{aligned}
\mathbf{u}_1^T\mathbf{u}_2 &= \mathbf{u}_1^T\tilde{\mathbf{u}}_2 - \left(\frac{\tilde{\mathbf{u}}_2^T\mathbf{u}_1}{||\mathbf{u}_1||}\right)\mathbf{u}_1^T\mathbf{u}_1 - \left(\frac{\tilde{\mathbf{u}}_2^T J\mathbf{u}_1}{||J\mathbf{u}_1||}\right)\mathbf{u}_1^T J\mathbf{u}_1\\
&= \mathbf{u}_1^T\tilde{\mathbf{u}}_2 - \tilde{\mathbf{u}}_2^T\mathbf{u}_1 = 0
\end{aligned} \tag{7}$$

$$\begin{aligned}
\omega(\mathbf{u}_1, \mathbf{u}_2) &= -\mathbf{u}_1^T J\mathbf{u}_2\\
&= -\mathbf{u}_1^T J\tilde{\mathbf{u}}_2 + \left(\frac{\tilde{\mathbf{u}}_2^T\mathbf{u}_1}{||\mathbf{u}_1||}\right)\mathbf{u}_1^T J\mathbf{u}_1\\
&\quad + \left(\frac{\tilde{\mathbf{u}}_2^T J\mathbf{u}_1}{||J\mathbf{u}_1||}\right)\mathbf{u}_1^T JJ\mathbf{u}_1\\
&= -\mathbf{u}_1^T J\tilde{\mathbf{u}}_2 - \mathbf{u}_2^T J\mathbf{u}_1\\
&= \omega\big(\mathbf{u}_1, \tilde{\mathbf{u}}_2\big) + \omega\big(\tilde{\mathbf{u}}_2, \mathbf{u}_1\big) = 0.
\end{aligned} \tag{8}$$

Finishing with $\mathbf{u}_1 \to \mathbf{u}_1/||\mathbf{u}_1||$ and $\mathbf{u}_2 \to \mathbf{u}_2/||\mathbf{u}_2||$, we have $\mathbf{u}_1^T\mathbf{u}_1 = \mathbf{u}_2^T\mathbf{u}_2 = 1$ and $\omega(\mathbf{u}_1, \mathbf{u}_1) = \omega(\mathbf{u}_2, \mathbf{u}_2) = 0$ by skew-symmetry of the symplectic 2-form. This completes the base step for our proof by induction. Moving to the induction step, we assume we have generated $j-1$ vectors that satisfy $\mathbf{u}_l^T\mathbf{u}_m = \delta_{lm}$ and $\omega(\mathbf{u}_l, \mathbf{u}_m) = 0,\ \forall l, m = 1, \ldots, j-1$. The next vector in the recursive algorithm is computed as $\mathbf{u}_j$ in (6). We then find that,

$$\begin{aligned}
\mathbf{u}_l^T\mathbf{u}_j &= \mathbf{u}_l^T\tilde{\mathbf{u}}_j\\
&\quad - \sum_{k=1}^{j-1}\left[\left(\frac{\tilde{\mathbf{u}}_j^T\mathbf{u}_k}{||\mathbf{u}_k||}\right)\mathbf{u}_l^T\mathbf{u}_k - \left(\frac{\tilde{\mathbf{u}}_j^T J\mathbf{u}_k}{||J\mathbf{u}_k||}\right)\mathbf{u}_l^T J\mathbf{u}_k\right]\\
&= \mathbf{u}_l^T\tilde{\mathbf{u}}_j - \tilde{\mathbf{u}}_j^T\mathbf{u}_l = 0
\end{aligned} \tag{9}$$

$$\begin{aligned}
\omega\big(\mathbf{u}_l, \mathbf{u}_j\big) &= -\mathbf{u}_l^T J\mathbf{u}_j\\
&= -\mathbf{u}_l^T J\tilde{\mathbf{u}}_j\\
&\quad + \sum_{k=1}^{j-1}\left[\left(\frac{\tilde{\mathbf{u}}_j^T\mathbf{u}_k}{||\mathbf{u}_k||}\right)\mathbf{u}_l^T J\mathbf{u}_k + \left(\frac{\tilde{\mathbf{u}}_j^T J\mathbf{u}_k}{||J\mathbf{u}_k||}\right)\mathbf{u}_l^T JJ\mathbf{u}_k\right]\\
&= -\mathbf{u}_l^T J\tilde{\mathbf{u}}_j - \mathbf{u}_j^T J\mathbf{u}_l\\
&= \omega\big(\mathbf{u}_l, \tilde{\mathbf{u}}_j\big) + \omega\big(\tilde{\mathbf{u}}_j, \mathbf{u}_l\big) = 0.
\end{aligned} \tag{10}$$

Finally taking $\mathbf{u}_j \to \mathbf{u}_j/||\mathbf{u}_j||$, we have $\mathbf{u}_j^T\mathbf{u}_j = 1$ and $\omega(\mathbf{u}_j, \mathbf{u}_j) = 0$. Hence, by induction, we have developed an n-dimensional basis $\{\mathbf{u}_i\}_{i=1}^n$ for a Lagrangian subspace of $\mathcal{T}_{z_0}\mathbb{R}^{2n}$ that satisfies $\mathbf{u}_i^T\mathbf{u}_j = \delta_{ij}$ and $\omega(\mathbf{u}_i, \mathbf{u}_j) = 0\ \forall i, j = 1, \ldots, n$. Next, we construct an orthogonal basis for the complement of the Lagrangian subspace by defining $\{\mathbf{v}_i\}_{i=1}^n$ where $\mathbf{v}_i = J\mathbf{u}_i$. By construction, these vectors satisfy $\mathbf{v}_i^T\mathbf{v}_j = \delta_{ij}$ and $\omega(\mathbf{v}_i, \mathbf{v}_j) = 0\ \forall i, j = 1, \ldots, n$. Putting both bases, we define $B = \{\mathbf{u}_1, \ldots, \mathbf{u}_n; \mathbf{v}_1, \ldots, \mathbf{v}_n\}$ as a basis for $\mathcal{T}_{z_0}\mathbb{R}^{2n}$. Note that,

$$\begin{aligned}
\mathbf{u}_i^T\mathbf{v}_j &= \mathbf{u}_i^T J\mathbf{u}_j = -\omega\big(\mathbf{u}_i, \mathbf{u}_j\big) = 0\\
\omega\big(\mathbf{u}_i, \mathbf{v}_j\big) &= -\mathbf{u}_i^T J\mathbf{v}_j = -\mathbf{u}_i^T JJ\mathbf{u}_j = \mathbf{u}_i^T\mathbf{u}_j = \delta_{ij}.
\end{aligned} \tag{11}$$

With (11), it is clear that the basis satisfies the conditions of (5) and that of an orthonormal basis. Hence, B is an orthosymplectic basis. We then choose $\mathbf{u}_1 = \tilde{\mathbf{u}}_1 = -J\widehat{\nabla_z\mathcal{H}}$, and, thus, $\mathbf{v}_1 = J\mathbf{u}_1 = \widehat{\nabla_z\mathcal{H}}$ where $\hat{\mathbf{a}} = \mathbf{a}/||\mathbf{a}||$. ■

From this theorem comes an important corollary that allows us to develop a linear, symplectic map for the tangent space using an orthosymplectic basis.

Corollary 1 (Symplectic Change of Basis Matrix): The change of basis matrix,

$$R(\mathbf{z}_0) = \begin{bmatrix} \mathbf{u}_1^T \\ \dots \\ \mathbf{u}_n^T \\ \mathbf{v}_1^T \\ \dots \\ \mathbf{v}_n^T \end{bmatrix} = \begin{bmatrix} -J\widehat{\nabla_z \mathcal{H}}^T \\ \dots \\ \mathbf{u}_n^T \\ \widehat{\nabla_z \mathcal{H}}^T \\ \dots \\ J\mathbf{u}_n^T \end{bmatrix} \in \mathbb{R}^{2n\times 2n} \tag{12}$$

is a symplectic matrix.

Proof: A symplectic matrix $\mathcal{A}$ satisfies $\mathcal{A}J\mathcal{A}^T = J$. Taking $\mathcal{A}$ to be $R(\mathbf{z}_0)$, (5) and (11) capture the matrix multiplication of $\mathcal{A}J\mathcal{A}^T = J$, decomposed in row-column multiplication. ■

Identifying the tangent space $\mathcal{T}_{z_0}\mathbb{R}^{2n}$ with its Euclidean representation ($\mathbb{R}^{2n}$), we transform $\delta\mathbf{z}_0 \in \mathcal{T}_{z_0}\mathbb{R}^{2n}$ to $R(\mathbf{z}_0)\delta\mathbf{z}_0$. Denoting $\xi = R(\mathbf{z}_0)\delta\mathbf{z}_0$, the new canonical coordinates spanning the transformed space are referred to as (Q_i, P_i). We identify ξ with its Euclidean representation $\xi = [Q_1, \dots, Q_n, P_1, \dots, P_n]$. In the vicinity of $\mathbf{z}_0$, these coordinates aid in the analysis of motion on iso-energetic contours foliating the symplectic manifold $(\mathbb{R}^{2n}, \omega)$. Fixing $P_1 = \widehat{\nabla_z \mathcal{H}}^T \delta\mathbf{z}_0 = 0$ restricts our study to neighbouring points $\mathbf{z}_0'$ at the same Hamiltonian value as $\mathbf{z}_0$.

Definition 6 (Energy-Surface Volume) Let Σ_E denote the iso-energetic set of all neighbouring states fixed to the Hamiltonian value $\mathcal{H}(\mathbf{z}_0)$. Since the system is autonomous, Σ_E is an invariant set. We can define the energy-surface volume ($2n$–1)-form,

$$\eta(\xi_1, \dots, \xi_{2n-1}) = \mathrm{d}Q_1 \wedge \mathrm{d}Q_2 \wedge \mathrm{d}P_2 \wedge \dots \\ \dots \wedge \mathrm{d}Q_n \wedge \mathrm{d}P_n(\xi_1, \dots, \xi_{2n-1}), \tag{13}$$

where $\xi_1, \dots, \xi_{2n-1} \in \mathcal{T}_{z_0}\mathbb{R}^{2n}$ are tangent vectors pointing in Σ_E. In coordinate representation, $\xi_1, \dots, \xi_{2n-1}$ form the sidelengths of a ($2n$–1)-dimensional cube stemming from $\mathbf{z}_0$, embedded entirely in Σ_E. This cube approximately moves at the same speed as the point $\mathbf{z}_0$ does on the symplectic manifold. We denote this speed with $v = ||-J\nabla_z\mathcal{H}(\mathbf{z}_0)|| = ||\nabla_z\mathcal{H}(\mathbf{z}_0)||$.

Theorem 2 (Iso-Energetic Cube Growth) The ($2n-1$)-dimensional volume η of a cube embedded within an iso-energetic manifold is directly proportional to the speed of the cube v.

Proof: Note that,

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{v}{\eta}\right) = \frac{\dot{v}}{\eta} - \frac{v\dot{\eta}}{\eta^2} = \frac{1}{\eta}\left(\dot{v} - \frac{v\dot{\eta}}{\eta}\right). \tag{14}$$

We identify the time derivative of the ($2n$-1)-form η with its Lie derivative. Let $\mathcal{L}_{\mathcal{H}}$ be the Lie derivative associated with the flow (ϕ_t) generated by the Hamiltonian $\mathcal{H}$ at $\mathbf{z}_0 \in \mathbb{R}^{2n}$. Then, $\dot{\eta} = \mathcal{L}_{\mathcal{H}}(\eta)$. The relevant properties of the Lie derivative include the following,

$$\mathcal{L}_{\mathcal{H}}(\mathrm{d}\alpha \wedge \mathrm{d}\beta) = (\mathcal{L}_{\mathcal{H}}(\mathrm{d}\alpha)) \wedge \mathrm{d}\beta + \mathrm{d}\alpha \wedge (\mathcal{L}_{\mathcal{H}}(\mathrm{d}\beta)) \\ \mathcal{L}_{\mathcal{H}}(\mathrm{d}\gamma) = \mathrm{d}(\mathcal{L}_{\mathcal{H}}(\gamma)) = \mathrm{d}\left(\frac{\mathrm{d}\gamma}{\mathrm{d}t}\right), \tag{15}$$

where $\mathrm{d}\alpha$, $\mathrm{d}\beta$, and $\mathrm{d}\gamma$ are differential forms [14].

Applying (15) to η,

$$\begin{aligned} \dot{\eta} &= \mathcal{L}_{\mathcal{H}}(\mathrm{d}Q_1 \wedge \mathrm{d}Q_2 \wedge \mathrm{d}P_2 \cdots \wedge \mathrm{d}Q_n \wedge \mathrm{d}P_n) \\ &= \mathrm{d}\left(\frac{\mathrm{d}Q_1}{\mathrm{d}t}\right) \wedge \dots \wedge \mathrm{d}P_n + \dots + \mathrm{d}Q_1 \wedge \dots \wedge \mathrm{d}\left(\frac{\mathrm{d}P_n}{\mathrm{d}t}\right) \\ &= \left(\frac{\partial}{\partial Q_1}\left(\frac{\mathrm{d}Q_1}{\mathrm{d}t}\right)\mathrm{d}Q_1 + \frac{\partial}{\partial P_1}\left(\frac{\mathrm{d}Q_1}{\mathrm{d}t}\right)\mathrm{d}P_1 + \dots \right. \\ &\quad \left. \dots + \frac{\partial}{\partial P_n}\left(\frac{\mathrm{d}Q_1}{\mathrm{d}t}\right)\mathrm{d}P_n\right) \wedge \dots \wedge \mathrm{d}P_n + \dots \\ &\quad \dots + \mathrm{d}Q_1 \wedge \dots \wedge \left(\frac{\partial}{\partial Q_1}\left(\frac{\mathrm{d}P_n}{\mathrm{d}t}\right)\mathrm{d}Q_1 \right. \\ &\quad \left. + \frac{\partial}{\partial P_1}\left(\frac{\mathrm{d}P_n}{\mathrm{d}t}\right)\mathrm{d}P_1 + \dots + \frac{\partial}{\partial P_n}\left(\frac{\mathrm{d}P_n}{\mathrm{d}t}\right)\mathrm{d}P_n\right). \end{aligned} \tag{16}$$

The infinitesimal cube spanned by tangent vectors $\xi_1, \dots, \xi_{2n-1}$ lies entirely in Σ_E. Then, all $\mathrm{d}P_1$ terms vanish. Any repeated differential, coordinate 1-forms also vanish such that $\mathrm{d}Q_i \wedge \dots \wedge \mathrm{d}Q_i \wedge \dots = 0$ and $\mathrm{d}P_i \wedge \dots \wedge \mathrm{d}P_i \wedge \dots = 0$ [11]. Equation (16) simplifies to,

$$\begin{aligned} \dot{\eta} &= \left[\frac{\partial}{\partial Q_1}\left(\frac{\mathrm{d}Q_1}{\mathrm{d}t}\right) + \dots + \frac{\partial}{\partial P_n}\left(\frac{\mathrm{d}P_n}{\mathrm{d}t}\right)\right]\eta. \\ &= \left[\sum_{i=1}^{n}\left[\frac{\partial}{\partial Q_i}\left(\frac{\mathrm{d}Q_i}{\mathrm{d}t}\right) + \frac{\partial}{\partial P_i}\left(\frac{\mathrm{d}P_i}{\mathrm{d}t}\right)\right] - \frac{\partial}{\partial P_1}\left(\frac{\mathrm{d}P_1}{\mathrm{d}t}\right)\right]\eta \\ &= \left[\nabla_\xi \cdot \dot{\xi} - \frac{\partial}{\partial P_1}\left(\frac{\mathrm{d}P_1}{\mathrm{d}t}\right)\right]\eta = -\frac{\partial}{\partial P_1}\left(\frac{\mathrm{d}P_1}{\mathrm{d}t}\right)\eta, \end{aligned} \tag{17}$$

where $\nabla_\xi \cdot \dot{\xi} = 0$ because (Q_i, P_i) is a volume-preserving, canonical coordinate set. Equation (14) reduces to,

$$\implies \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{v}{\eta}\right) = \frac{1}{\eta}\left(\dot{v} + v\left(\frac{\partial}{\partial P_1}\left(\frac{\mathrm{d}P_1}{\mathrm{d}t}\right)\right)\right). \tag{18}$$

Looking at the first term of (18), we have,

$$\dot{v} = \frac{\mathrm{d}}{\mathrm{d}t}(||\nabla_z\mathcal{H}||) = -\frac{1}{||\nabla_z\mathcal{H}||}\nabla_z\mathcal{H}^T\nabla_{zz}\mathcal{H}J\nabla_z\mathcal{H}. \tag{19}$$

The evolution of $\delta\mathbf{z}_0$ is given by (2). Examining the second term of (18), we find,

$$\begin{aligned} \frac{\mathrm{d}P_1}{\mathrm{d}t} &= \dot{P}_1 = \left[\frac{\mathrm{d}}{\mathrm{d}t}(\widehat{\nabla_z\mathcal{H}})^T\mathbb{I}_{2n\times 2n} - \widehat{\nabla_z\mathcal{H}}^T J\nabla_{zz}\mathcal{H}\right]\delta\mathbf{z}_0 \\ \frac{\partial\dot{P}_1}{\partial P_1} &= \left[\frac{\mathrm{d}}{\mathrm{d}t}(\widehat{\nabla_z\mathcal{H}})^T\mathbb{I}_{2n\times 2n} - \widehat{\nabla_z\mathcal{H}}^T J\nabla_{zz}\mathcal{H}\right]\frac{\partial\delta\mathbf{z}_0}{\partial P_1}, \\ &\text{where } P_1 = \widehat{\nabla_z\mathcal{H}(\mathbf{z}_0)}^T\delta\mathbf{z}_0 \longrightarrow \frac{\partial\delta\mathbf{z}_0}{\partial P_1} = \widehat{\nabla_z\mathcal{H}}. \\ \rightarrow \frac{\partial\dot{P}_1}{\partial P_1} &= \frac{\mathrm{d}}{\mathrm{d}t}(\widehat{\nabla_z\mathcal{H}})^T\widehat{\nabla_z\mathcal{H}} - \widehat{\nabla_z\mathcal{H}}^T J\nabla_{zz}\mathcal{H}\widehat{\nabla_z\mathcal{H}} \end{aligned} \tag{20}$$

The first scalar term in (20) vanishes because any unit vector $\hat{\mathbf{a}}$ satisfies $\frac{\mathrm{d}}{\mathrm{d}t}(1) = \frac{\mathrm{d}}{\mathrm{d}t}(\hat{\mathbf{a}}^T\hat{\mathbf{a}}) = 2\frac{\mathrm{d}}{\mathrm{d}t}(\hat{\mathbf{a}})^T\hat{\mathbf{a}} = 0$. The second scalar term in (20) is invariant under the transpose operator. With $v = ||\nabla_z\mathcal{H}||$, (20) is used to write,

$$v\left(\frac{\partial\dot{P}_1}{\partial P_1}\right) = \frac{1}{||\nabla_z\mathcal{H}||}\nabla_z\mathcal{H}^T\nabla_{zz}\mathcal{H}J\nabla_z\mathcal{H}. \tag{21}$$

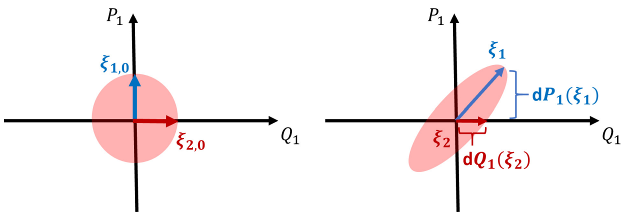


Fig. 1. A cartoon depicting the volume widths, $dP_1(\xi_1)$ and $dQ_1(\xi_2)$, governed by (24) and (29).

Then, the first and second terms of (18), listed in (19) and (21), cancel, leaving us with,

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{v}{\eta}\right) = 0 \implies \eta \propto v. \tag{22}$$

To address the problem in Section III, we use the behaviour of η and other volume-like symplectic invariants. ∎

Theorem 3 (Decoupled Phase Space Stretching) The flexing of phase space along the $-J\widehat{\nabla_z \mathcal{H}}(\mathbf{z}_0)$ and $\widehat{\nabla_z \mathcal{H}}(\mathbf{z}_0)$ directions is analytically predictable and decoupled from other basis directions.

Proof: Consider a $2n$-dimensional volume element spanned by the vectors $\xi_1, \ldots, \xi_{2n} \in \mathcal{T}_{z_0}\mathbb{R}^{2n}$. Without loss of generality, we assume ξ_1 has a non-zero component along $\widehat{\nabla_z \mathcal{H}}(\mathbf{z}_0)$ and $\xi_2, \ldots, \xi_{2n}$ fall along $-J\widehat{\nabla_z \mathcal{H}}(\mathbf{z}_0), \mathbf{u}_2, \mathbf{v}_2, \ldots, \mathbf{u}_n, \mathbf{v}_n$, respectively. The volume contained within this infinitesimal element is given by,

$$\begin{aligned} \mathrm{d}V &= \mathrm{d}P_1 \wedge \mathrm{d}Q_1 \wedge \cdots \wedge \mathrm{d}P_n \wedge \mathrm{d}Q_n(\xi_1, \ldots \xi_{2n}) \\ &= \mathrm{d}P_1(\xi_1) \cdot \eta(\xi_2, \ldots, \xi_{2n}), \end{aligned} \tag{23}$$

where exterior algebra properties have been used to express the $2n$-form $\mathrm{d}V$ as a product of the 1-form $\mathrm{d}P_1$ and the ($2n$-1)-form η [11]. Noting that $\mathrm{d}V$ is preserved by Liouville's Theorem, we find from (22) and (23),

$$dP_1(\xi_1) \propto ||\nabla_z \mathcal{H}||^{-1}, \tag{24}$$

for any vector ξ_1 initially with a non-zero $\widehat{\nabla_z \mathcal{H}}(\mathbf{z}_0)$ component. Considering the ($2n$-1)-form η again, we can leverage exterior algebra properties to write,

$$\begin{aligned} &\eta(\xi_2, \ldots, \xi_{2n}) \\ &\quad = \mathrm{d}Q_1(\xi_2) \cdot \mathrm{d}Q_2 \wedge \mathrm{d}P_2 \wedge \cdots \wedge \mathrm{d}Q_n \wedge \mathrm{d}P_n(\xi_3, \ldots, \xi_{2n}). \end{aligned} \tag{25}$$

The tangent vectors $\xi_3, \ldots, \xi_{2n}$ span a ($2n$–2)-dimensional cube. Associated with this cube is an invariant ($2n$–2)-form $\omega^{(2n-2)}$ where,

$$\begin{aligned} \omega^{(2n-2)}(\xi_3, \ldots, \xi_{2n}) = \sum_{1 \le i_1 \le \cdots \le i_{n-1} \le n} \Big(&\mathrm{d}Q_{i_1} \wedge \mathrm{d}P_{i_1} \wedge \ldots \\ &\cdots \wedge \mathrm{d}Q_{i_{n-1}} \wedge \mathrm{d}P_{i_{n-1}}(\xi_3, \ldots, \xi_{2n}) \Big), \end{aligned} \tag{26}$$

is the higher dimensional generalization of the standard symplectic 2-form ω [11]. Note that $\xi_3, \ldots, \xi_{2n} \in \mathcal{T}_{z_0}\mathbb{R}^{2n}$ are orthogonal to $\widehat{\nabla_z \mathcal{H}}(\mathbf{z}_0)$. Therefore, $\mathrm{d}P_1(\xi_i) = 0 \ \forall i = 3, \ldots, 2n$. All terms in the summation of (26) vanish apart from the one excluding the $\mathrm{d}P_1$. Then, (26) reduces to,

$$\begin{aligned} \omega^{(2n-2)}(\xi_3, \ldots, \xi_{2n}) = \mathrm{d}Q_2 \wedge \mathrm{d}P_2 \wedge \cdots \wedge \mathrm{d}Q_n \wedge \mathrm{d}P_n(\xi_3, \\ \ldots, \xi_{2n}). \end{aligned} \tag{27}$$

Equation (25) can be rewritten as,

$$\eta(\xi_2, \ldots, \xi_{2n}) = \mathrm{d}Q_1(\xi_2) \cdot \omega^{(2n-2)}(\xi_3, \ldots, \xi_{2n}). \tag{28}$$

Since $\omega^{(2n-2)}(\xi_3, \ldots, \xi_{2n})$ is an invariant and $\eta \propto ||\nabla_z \mathcal{H}(\mathbf{z}_0)||$, we have,

$$dQ_1(\xi_2) \propto ||\nabla_z \mathcal{H}||, \tag{29}$$

for any tangent vector ξ_2 pointing along $-J\widehat{\nabla_z \mathcal{H}}(\mathbf{z}_0)$. Equations (24) and (29) indicate how phase space stretches along the $\widehat{\nabla_z \mathcal{H}}(\mathbf{z}_0)$ and $-J\widehat{\nabla_z \mathcal{H}}(\mathbf{z}_0)$ directions, indicated entirely by the magnitude of the local gradient of the Hamiltonian. This is captured in Fig. 1, showing a volume and its widths at two times. From the figure, control strategies collapsing the volume along $\widehat{\nabla_z \mathcal{H}}(\mathbf{z}_0)$, i.e., the P_1-axis, at the second time will require more effort due to the larger state discontinuities. Pinching of ξ_2 along the $-J\widehat{\nabla_z \mathcal{H}}(\mathbf{z}_0)$ direction is compensated by stretching of ξ_1 along the $\widehat{\nabla_z \mathcal{H}}(\mathbf{z}_0)$ direction because the proportionality factors are reciprocally related. Then, stretching and pinching in the remaining ($2n$-2)-dimensional subspace acts to preserve $\omega^{(2n-2)}$. Hence, there is a decoupling of the tangent space $\mathcal{T}_{z_0}\mathbb{R}^{2n}$ into a 2-dimensional subspace and ($2n$-2)-dimensional subspace that preserve their respective measures of volume. ∎

V. Numerical Example

A. Two-Body Problem

Consider the mutual gravitational attraction between two point masses. The canonical coordinates are $\mathbf{z} = (q_1, q_2, p_1, p_2) = (x, y, p_x, p_y)$, representing mutual separation between the two masses in Cartesian positions and momenta. The scalar Hamiltonian tied to the two-body problem (2BP) is given as,

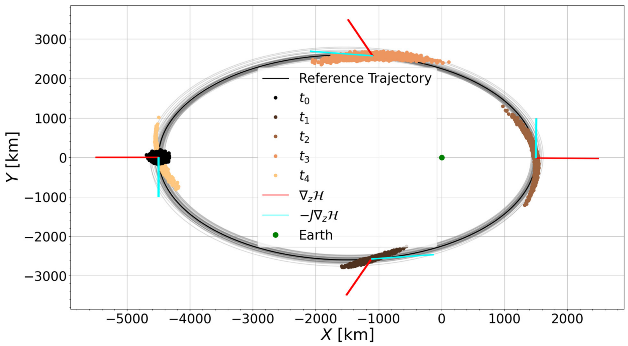
$$\mathcal{H} = \frac{1}{2}p^2 - \frac{\mu}{r}, \tag{30}$$

where $p = \sqrt{p_x^2 + p_y^2}$, $r = \sqrt{x^2 + y^2}$, and μ is the reduced mass parameter. The associated equations of motion are,

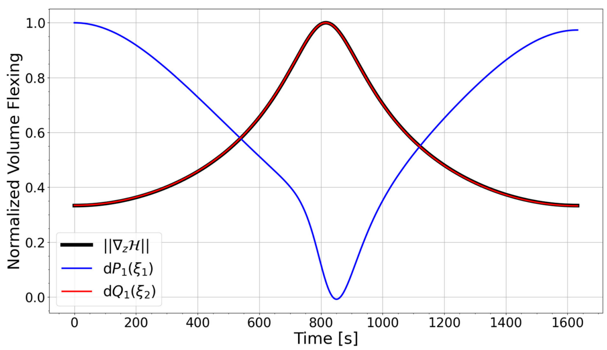
$$\frac{\mathrm{d}\mathbf{z}}{\mathrm{d}t} = -J\nabla_z \mathcal{H}(\mathbf{z}) = \left(p_x, p_y, -\frac{\mu x}{r^3}, -\frac{\mu y}{r^3}\right). \tag{31}$$

We set $\mu = 4 \times 10$ km^3/s^2 to capture the Earth two-body problem. Note that (31) defines the local flow vector and the local gradient vector is given as $\nabla_z \mathcal{H} = (\mu x/r^3, \mu y/r^3, p_x, p_y)$.

We simulate a collection of states within a 4-dimensional volume sampled from a multivariate, Gaussian distribution with a mean eccentric orbit $\bar{\mathbf{z}}_0 = (-4500 \text{ km}, 0 \text{ km}, 0 \text{ km}, -6.\bar{6} \text{ km/s})$ and a covariance $P_0 = \mathrm{diag}([2500 \text{ km}^2, 2500 \text{ km}^2, 10^{-4} \text{ km}^2/\text{s}^2, 10^{-4} \text{ km}^2/\text{s}^2])$. Fig. 2(a) captures the position-space (q_1, q_2) evolution of our volume at four different times spaced about a close passage of the Earth.



(a) Nonlinearly mapped volume evolution in phase space.



(b) Time history of nonlinearly mapped volume widths, $\mathrm{d}P_1(\xi_1)$ and $\mathrm{d}Q_1(\xi_2)$, normalized to their maximum value.

Fig. 2. Phase space flexing about a reference trajectory in the two-body problem.

The position-space projections of the local flow and gradient directions are also shown in red and blue, respectively. In position-space, the volume is stretched along $-J\widehat{\nabla_z\mathcal{H}}(\mathbf{z}_0)$ and pinched along $\widehat{\nabla_z\mathcal{H}}(\mathbf{z}_0)$ upon close approach with the Earth. Fig. 2(b) captures the volume widths, $\mathrm{d}P_1(\xi_1)$ and $\mathrm{d}Q_1(\xi_2)$. The blue curve in Fig. 2(b) slightly deviates from a reciprocal relationship with the black curve. However, our simulated volume is not infinitesimal and will not have perfect agreement. Nonetheless, the widths $\mathrm{d}P_1(\xi_1)$ and $\mathrm{d}Q_1(\xi_2)$ numerically computed are approximately consistent with (24) and (29). To collapse a phase space volume and place a spacecraft onto a target orbit, we expect fuel-minimizing control strategies to respond by collapsing the volume primarily along $\widehat{\nabla_z\mathcal{H}}$ through periapsis passage and then along $-J\widehat{\nabla_z\mathcal{H}}$ through apoapsis passage.

VI. Conclusion

For autonomous Hamiltonian systems, we have distilled the phase space in the neighbourhood of a reference state into two decoupled, lower-dimensional subspaces of dimensions 2 and $2n$-2, respectively. Flexing of state deviations within the 2-dimensional subspace occur along the local flow and gradient direction, and is entirely indicated by the magnitude of the Hamiltonian gradient. This result is shown using natural invariants arising from the symplectic structure in Hamiltonian systems, along with the algebraic properties of exterior algebraic forms. Specifically, an orthosymplectic basis for the tangent space is constructed that characterizes measures of size within an iso-energetic surface, leading to flexing constraints. The result is demonstrated in the two-body problem. The flexing forecasts inform the behaviour of fuel-minimizing control strategies that leverage natural volume deformations to their benefit. Future work aims to interpret these flexing forecasts in the extended phase space tied to higher dimensional optimal control problems.

References

[1] E. L. Jenson and D. J. Scheeres, "Multi-objective optimization of covariance and energy for asteroid transfers," *J. Guid. Control Dyn.*, vol. 44, no. 7, pp. 1253–1265, 2021.
[2] D. J. Dichmann, C. M. Alberding, and W. Y. Yu, "Stationkeeping monte carlo simulation for the James Web space telescope," in *Proc. 26th Int. Symp. Space Flight Dyn.*, 2014, pp. 1–21.
[3] H. Goldstein, C. Poole, and J. Safko, *Classical Mechanics*. Boston, MA, USA: Addison-Wesley, 2002, pp. 388–396.
[4] L. S. Pontryagin, *Mathematical Theory of Optimal Processes, Classics of Soviet Mathematics*. Abingdon, U.K.: Taylor & Francis, 1987.
[5] O. P. Agrawal and D. Baleanu, "A hamiltonian formulation and a direct numerical scheme for fractional optimal control problems," *J. Vibr. Control*, vol. 13, nos. 9–10, pp. 1269–1281, 2007.
[6] N. Javanmardi, A. Yaghmaei, and M. J. Yazdanpanah, "Spacecraft formation flying in the port-Hamiltonian framework," *Nonlin. Dyn.*, vol. 99, no. 4, pp. 2765–2783, 2020.
[7] A. van der Schaft, "Port-Hamiltonian systems: An introductory survey," in *Proc. Int. Cong. Math. Vol. III, Euro. Math. Society Pub. House (EMS Ph)*, 2006, pp. 1339–1365.
[8] I. Goldhirsch, P.-L. Sulem, and S. A. Orszag, "Stability and Lyapunov stability of dynamical systems: A differential approach and a numerical method," *Physica D, Nonlin. Phen.*, vol. 27, no. 3, pp. 311–337, 1987.
[9] D. R. Smith, *An Introduction To Continuum Mechanics After Truesdell and Noll*. Dordrech, The Netherlands: Springer, 1993, p. 111.
[10] C. R. Short, D. Blazevski, K. C. Howell, and G. Haller, "Stretching in phase space and applications in general nonautonomous multi-body problems," *Cel. Mech. Dyn. Astro.*, vol. 122, pp. 213–238, Jul. 2015.
[11] V. I. Arnold, *Mathematical Methods of Classical Mechanics*. New York, NY, USA: Springer, 1989, pp. 163–192.
[12] V. Guibout, "The Hamilton-Jacobi theory for solving two-point boundary value problems: Theory and numerics with application to spacecraft formation flight, optimal control and the study of phase space structure," Ph.D. dissertation, Dept. Aerosp. Eng., Univ. Michigan, Ann Arbor, MI, USA, 2004, pp. 237–239.
[13] M. de Gosson and F. Luef, "Symplectic capacities and the geometry of uncertainty: The irruption of symplectic topology in classical and quantum mechanics," *Phys. Rep.*, vol. 484, no. 5, pp. 131–179, 2009.
[14] J. M. Lee, *Introduction to Smooth Manifolds*. New York, NY, USA: Springer, 2003, pp. 372–373.